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NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High Section
Preliminary Examination
Higher 2

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| CANDIDATE NAME | |

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| BIOLOGY CLASS | | REGISTRATION NUMBER | |

# BIOLOGY 9747/03

**Paper 3 Applications Paper**

**September 2009**

**1 hour 30 minutes**

**Additional Materials: Answer Paper**

**READ THESE INSTRUCTIONS FIRST**

Write your Biology class, registration number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| 1 | / 15 |
| 2 | / 15 |
| 3 | / 15 |
| 4 | / 20 |
| **TOTAL** | **/ 65** |

Answer **all** questions.

**1** Growth hormone (GH) deficient children can be treated with human GH that is manufactured by recombinant DNA technology.

Fig. 1.1 outlines the procedures for cloning human GH cDNA in a DNA plasmid. Suitable recipient bacteria are then treated to form bacteria with recombinant plasmids.

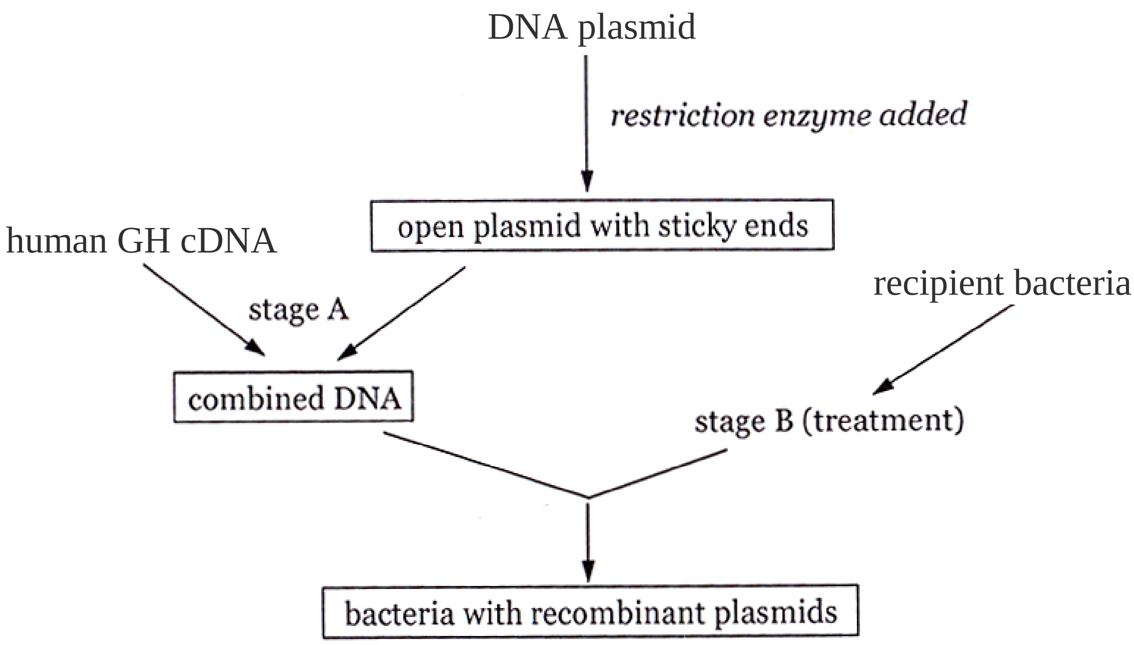


**Fig. 1.1**

**(a)** State **one** key property of the DNA plasmid that allows it to be used as a DNA cloning vector. [1]

- ***one origin of replication / at least one selectable marker / at least one unique restriction site***

**(b)** Explain why it is advantageous to use a restriction enzyme that produces an open plasmid with sticky ends. [2]

- ***The sticky ends of the open plasmid can form hydrogen-bonded base pairs (temporary associations) with complementary sticky ends of the target DNA (human GH cDNA) that is produced by the same restriction enzyme***
- ***Such temporary associations between the sticky ends greatly increase the chance of forming recombinant plasmids with the help of DNA ligase***

**(c)** Outline the steps involved in obtaining human GH cDNA. [3]

- ***Extract total mRNA from the cells of human pituitary gland***
- ***Convert the mRNAs to cDNAs using reverse transcriptase and DNA polymerase***
- ***Amplify only the human GH cDNA (flanked with a desired restriction site) using Polymerase Chain Reaction (PCR)***

**(d)** Name the enzyme used in stage A to form the combined DNA. [1]

- ***DNA ligase***

**(e)** Explain the need for suitable recipient bacteria to go through stage B before bacteria with recombinant plasmids are formed. [2]

- ***Not all bacteria undergo natural transformation***
- ***DNA, being negatively charged is a very hydrophilic molecule that will not normally pass through the hydrophobic cell membrane***
- ***There is thus a need for bacterial cells to be made "competent" to take up DNA***
- ***ref to chemical transformation or electroporation***

**(f)** Describe how bacteria can be screened for the presence of recombinant plasmids based on β-galactosidase activity. [4]

- ***Bacterial plasmids or Expression / Cloning vectors, each of which contains 1 antibiotic resistance gene (e.g. Amp $^r$) and unique cloning sites within the lacZ gene are used to transform bacteria that lack R plasmids***
- ***The bacteria are then plated on an agar plate containing ampicillin to eliminate those with no uptake of the cloning vectors***
- ***The same agar plate also contains X-Gal and IPTG whereby IPTG induces the production of functional β-galactosidase that catalyses the hydrolysis of X-Gal to a blue product***
- ***Clones of bacteria that contain the gene of interest will appear white while those that do not contain the gene of interest will appear blue***

**(g)** Following the identification of bacteria with recombinant plasmids, there was an attempt to induce the expression of the human GH cDNA. However, no functional human GH was obtained.

Suggest **two** reasons for this. [2]

- ***Human GH cDNA cannot be transcribed / expressed by the host organism (bacteria) since it was inserted into a cloning vector that does not have a promoter***

- ***mRNA that was produced from the transcription of human GH cDNA does not code for functional human GH since the cDNA was inserted into the expression vector in the wrong orientation***

[Total: 15]

**2** A genetic test was performed on DNA from two individuals, **P** and **Q**, to find the base sequence of a small part of the gene involved in cystic fibrosis (CF).

The different base sequences of **P** and **Q** are shown diagrammatically in Fig. 2.1.

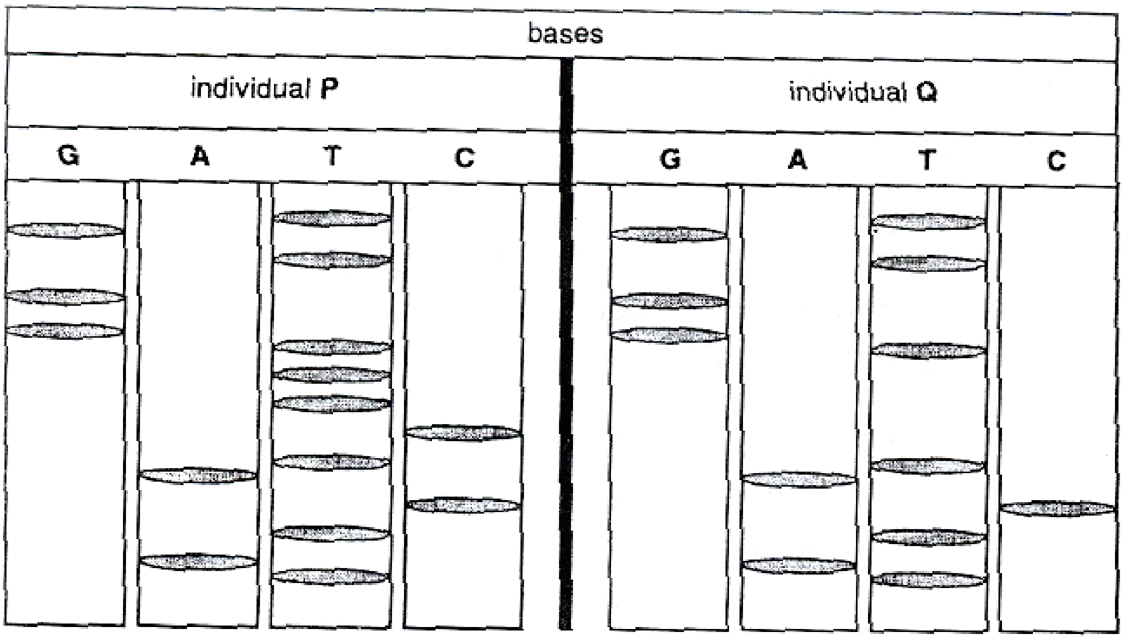


**Fig. 2.1**

**(a)** With reference to Fig. 2.1,

**(i)** identify how the base sequence of individual **Q** differs from that of **P**. [1]

- ***Deletion of three nucleotides / one codon***
- ***Identification of base sequence deleted – CTT***

**(ii)** explain the significance of the mutation identified in (a)(i). [3]

- ***Loss of amino acid phenylalanine***
- ***Defective CFTR protein***
- ***Defective CFTR protein will be retained in the ER will not reach the plasma membrane at all***
- ***Fewer / Absence of CFTR proteins in the membrane results in an overall decreased permeability of the cells to chloride ions***

**(b)** **(i)** Briefly describe the symptoms of CF in humans. [2]

- ***Thick and dehydrated mucus in lungs / gut ; Breathing difficulties***
- ***Prone to chronic infection / high level of lung infection***
- ***Presence of scar tissue (fibrosis) in the lungs***
- ***Prevents digestive enzymes from reaching intestines / blocks secretion of digestive enzymes from the pancreas***

**(ii)** Explain how CF is inherited. [2]

- ***Inherited via recessive allele***
- ***Carried on chromosome 7 (i.e. autosomal)***
- ***Heterozygotes are carriers***
- ***Sufferers are homozygous recessive***
- ***If both parents are carriers, there is a 1 in 4 chance that a child will have the disease***

Researchers are currently exploring the possibility of using adult stem cells from bone marrow, referred to as mesenchymal stem cells (MSCs), in therapy for CF.

**(c)** **(i)** Give **two** reasons for using MSCs in the treatment of CF. [2]

- ***1. Immune rejection is unlikely***
- ***Tissues grown are immunologically compatible with the person from whom the cells are harvested***
- ***2. Capable of limited differentiation***
- ***To yield the specialized cell types of the tissue from which it originated / lung epithelial cell***

**(ii)** Propose, using an appropriate gene delivery system, how MSCs can be used to treat CF. [3]

- *Ex vivo gene therapy*
- *Isolation of the normal CFTR gene*
- *Isolation of MSCs from bone marrow*
- *Introduce the normal CFTR gene into a viral vector (i.e. adenoviral vector)*
- *MSCs from patients were transduced with the viral vector*
- *Ex vivo expansion of culture*
- *Selection for expression of CFTR / Differentiate into lung epithelial cell*
- *Re-introduced the lung epithelial cells into patient's body*
- *Where they express the functional allele and synthesize CFTR protein*

**OR**

- *Ex vivo gene therapy*
- *Isolation of the normal CFTR gene*
- *Isolation of MSCs from bone marrow*
- *Clone the normal CFTR gene into a non-viral vector (i.e. cationic liposome)*
- *MSCs from patients were transfected with the non-viral vector*
- *Ex vivo expansion of culture*
- *Selection for expression of CFTR / Differentiate into lung epithelial cell*
- *Re-introduced the lung epithelial cells into patient's body*
- *Where they express the functional allele and synthesize CFTR protein*

**(iii)** Suggest **one** advantage and **one** disadvantage of using the gene delivery system stated in (c)(ii). [2]

***Based on viral gene delivery system***

- ***Advantage: High transduction efficiency***
- ***Disadvantage: Risk of immune response***

***Based on non-viral gene delivery system***

- ***Advantage: Not likely to generate immune response in patient's body***
- ***Disadvantage: Low transfection efficiency***

[Total: 15]

**3** Dinitroaniline herbicides are used for the selective control of grassy weeds amongst crops such as maize. These herbicides act by binding to the protein tubulin, thereby disrupting the formation of the spindle fibres and other cellular microtubules. Grasses resistant to these herbicides were found to have a mutation in the gene for tubulin. However, these altered tubulin proteins were still functional.

An experiment was performed on maize cells to see whether the mutation in the tubulin gene found in grass was actually responsible for dinitroaniline resistance. Maize cells were grown in tissue culture and divided into four groups (**M2**, **M1**, **N** and **C**) which were treated as follows:

**M2** was genetically engineered to express two copies of the mutated tubulin gene from grass.

**M1** was genetically engineered to express one copy of the mutated tubulin gene from grass.

**N** was genetically engineered to express the normal tubulin gene from grass.

**C** was not genetically engineered.

Each group of cells was then grown into a callus.

**(a)** **(i)** Briefly describe how plant cells may be grown to form many calli. [2]

- ***Grow plant cells in medium containing nutrients and agar with equal ratio of auxin and cytokinin***
- ***Plant cells will divide via mitosis***
- ***A callus may be subcultured to grow many calli***

**(ii)** State why the calli are grown in sterile conditions. [2]

- ***Prevent the growth of fungi or bacteria***
- ***So that resultant plants / plantlets are disease free***
- ***Calli are genetically identical to each other / clones***
- ***Prevent them from being wiped out by a disease***

Equal quantities of each maize callus were then grown in the presence of different concentrations of dinitroaniline herbicide. The results are shown in Fig. 3.1 in which the shaded circles show the relative size of each piece of callus tissue after growth in the presence of herbicides.

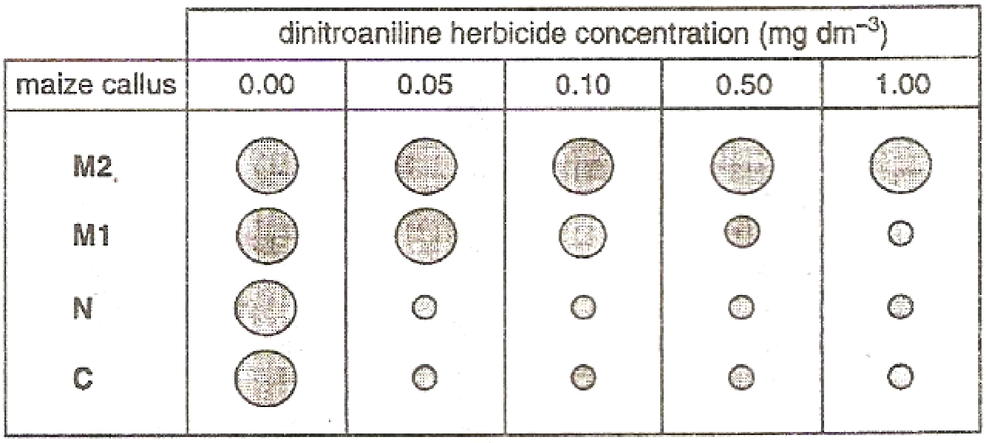


**Fig. 3.1**

**(b)** **(i)** Explain the purpose of maize calli **N** and **C** in this investigation. [2]

- *Controls*
- *Serves as a basis of comparison for M1 & M2*
- *Maize calli N and C grew minimally / grew the least / did not grow at all concentrations of herbicide tested OR any ref to results for N & C*
- *Shows that results in M1 & M2 are due to the presence of mutant tubulin gene from grass*
- *And not due to another factor / such as a presence of a foreign gene in the maize*
- *Also shows that the unmutated tubulin gene from grass cannot cause herbicide resistance*

**(ii)** Describe the evidence that the mutated tubulin gene from grass is responsible for dinitroaniline resistance. [3]

- ***Calli for M1 & M2 are larger than that for controls (N&C) at all concentrations of herbicide tested except at 1.00mg/dm-3 where M1 has a callus of the same size as controls***
- ***Herbicide has no effect on growth of M2 for all conc. tested / M2 callus grows to max size at all herbicide concs tested***
- ***At 0.1mg/dm-3 and beyond the higher the herbicide conc. the greater the inhibition on callus growth of M1 /***
- ***The conc. of herbicide that the cells can withstand is dependent / increases with the number of copies of the mutated tubulin gene present***
- ***M1 that has only one copy of the mutated tubulin gene is only resistant to the herbicide up to concentrations of 0.05mg/dm-3 / up to a low conc. only***
- ***M2 that has two copies of the mutated tubulin gene is resistant even up to the highest conc. of herbicide tested / 1.00mg/dm-3***
- ***Presence of mutated tubulin gene causes herbicide resistance (regardless of no. of copies) at low conc. of herbicide / 0.05mg/dm-3***

**(c)** A company did a study to determine the optimal concentration of dinitroaniline herbicide for use in increasing the yield of two groups of plant seedlings:

- genetically modified seedlings containing the mutated tubulin gene, and
- non-transformed seedlings.

Equal numbers of each type of seedlings were treated with varying concentrations of the herbicide.

Fig. 3.2 shows the percentage of each type of seedlings that grow under the different herbicide concentrations.

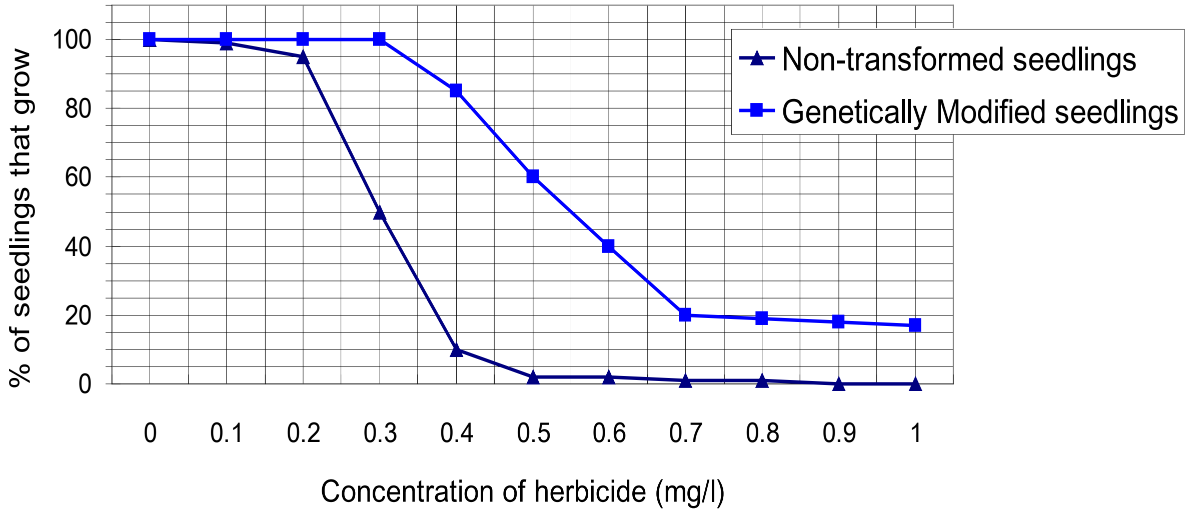


**Fig. 3.2**

With reference to Fig. 3.2,

**(i)** describe the effects of increasing concentration of herbicide on the **two** groups of seedlings. [3]

- ***At very low conc. of herbicide % of seedlings that grow for both non-transformed seedlings and GM seedlings are almost 100%***
- ***Conc. of 0-0.3mg/l of herbicide have no effect on GM seedling growth / % of GM seedling growth is maximum / 100%***
- ***From 0.2-0.4mg/l conc of herbicide % of non-transformed seedlings that grow drop sharply / greatly from 98% to 10%***
- ***From 0.2-0.4mg/l conc of herbicide % of GM seedlings that grow drops a little from 100% to 85%***
- ***At conc. of pesticide more than 0.4mg/l % growth of non-transformed seedlings continues to decrease from 10% to zero***
- ***From 0.3-0.7mg/l conc of herbicide & of GM seedlings that grow drops from 100% to 20%***
- ***Above 0.7mg/, increasing conc of herbicide does not result in any further / any more decrease in % of GM seedlings that grow / % of GM seedlings that grow plateaus***
- ***Effects of conc above 1.00mg/l were not tested / unknown***

**(ii)** suggest what the results for the non-transformed seedlings could show, other than serving as a control. [1]

- ***Shows possible effect of dinitroaniline herbicide on weeds, which have not been genetically modified / do not have the herbicide resistance gene***

**(iii)** explain why a farmer, looking at the results of the study, may prefer to use 0.4 mg/l of herbicide on his genetically modified seedlings instead of 0.3 mg/l. [2]

- ***Kills almost all of the non-transformed seedlings / less non-transformed seedlings survive at 0.4mg/l than 0.3mg/l. (statement of comparison)***
- ***Only 10% of the non-transformed seedlings survive at 0.4mg/l but at 0.3mg/l where almost 50% of non-transformed seedlings still survive (data comparison)***
- ***Quite a large % of GM seedlings, 80% can grow at 0.4mg/l***
- ***Gives the farmer a sufficient yield of GM crop***
- ***Less non-transformed seedlings means less competition for nutrients with GM crop***

***OR (link to weeds)***

- ***0.4mg/l is likely to kill a larger percentage of the weeds than 0.3mg/l (link to weeds)***
- ***(Less likelihood of weed population evolving herbicide resistance)***

[Total: 15]

Write your answers to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections (a), (b), etc., as indicated in the question.

**4(a)** Explain how RFLP analysis facilitated the process of DNA fingerprinting and suggest how it can be used to convict or exonerate a suspect of murder. [8]

- ***obtain DNA samples from (1) suspect and (2) crime scene DNA sample / evidence***
- ***amplify the amount of DNA to generate sufficient DNA for subsequent steps by PCR***
- ***digest both DNA samples with the same restriction enzyme(s)***
- ***differences in DNA sequence of both samples results in differences in restriction enzyme recognition site***
- ***(result in) different sizes / lengths / different numbers of restriction / DNA fragments***
- ***DNA restriction fragments are separated according to size / molecular weight by (agarose) gel electrophoresis***
- ***gel placed in NaOH to separate DNA into single-stranded DNA***
- ***(ssDNA) transferred onto nitrocellulose blot (by capillary action)***
- ***blot incubated with radioactively labeled nucleic acid probes which will hybridize / anneal to specific target nucleotide sequences by complementary base pairing***
- ***autoradiography carried out / blot and X-ray film exposed to X-ray***
- ***(target) DNA bands bound to radioactively labeled probes will show up as a black bands on the X-ray film***
- ***the resulting restriction fragment length polymorphism (RFLP) patterns observed are the DNA / genetic fingerprints of the suspect and crime scene DNA sample / evidence***
- ***By comparing the DNA bands in the DNA / genetic fingerprint of the suspect and the crime scene sample / evidence, one can see whether the DNA / genetic fingerprint of the suspect was similar enough to be closely related to the DNA / genetic fingerprint of the crime scene DNA sample***

**(b)** Explain how PCR facilitated the process of genetic diseases detection. [8]

- ***known DNA sequence of both the normal and mutant alleles / disease gene locus***
- ***ref to differences in short tandem repeats (STR) / variable number of tandem repeats (VNTRs) / differences in the nucleotide sequences of the alleles***
- ***(some) information about the normal and mutant / disease alleles used to design primers***
- ***PCR allows (these) specific gene sequences to be amplified***

- ***ref to obtaining DNA samples from normal individuals and (potential) disease carriers / patients (for comparison)***

***Process of PCR involves:***

- ***denaturation stage***
- ***heating to 95°C (accept valid range)***
- ***to separate DNA into single-stranded DNA by breaking the H bonds between them***
- ***annealing stage***
- ***cooling to 65°C (accept valid range)***
- ***forward and reverse / 2 different primers used***
- ***will anneal by complementary base pairing to DNA sequences located just outside the STR / target gene sequence***
- ***elongation stage***
- ***raising temperature to 72°C (accept valid range)***
- ***Taq / heat-stable DNA polymerase copy template / target DNA sequence***
- ***by adding complementary nucleotides to the 3'-OH ends on both primers***

- ***PCR cycle is repeated in an automated thermal cycler, resulting in rapid amplification of the target DNA sequence***

- ***the products of PCR are separated by (agarose) gel electrophoresis***
- ***amplified products of different STR alleles / normal and mutant alleles appear as distinct bands on the gel***
- ***ref to more STRs / VNTRs / repeats producing increased length of amplified PCR fragment / higher position of DNA band on gel***
- ***a genetic fingerprint of each individual is obtained***
- ***(genetic fingerprints of various individuals) can be compared and used for (named) disease detection***
- ***if the genetic fingerprint of a person share similar band(s) with that of a person definitive for the disease, one can conclude he/she will too suffer from the (named) disease***

**(c)** Discuss the social and ethical considerations for the use of gene therapy. [4]

***Social considerations***

- ***gene therapy is currently very expensive***
- ***concerns about gene therapy becoming accessible only to the wealthy***
- ***and possible genetic enhancements creating an advantage for those who can afford the treatment***

- ***a large number of gene therapy trials are currently directed to the treatment of diseases with an enormous social impact, such as cancer, muscular dystrophy, SCID and cystic fibrosis etc.***
- ***gene therapy might provide new forms of treatments directed to currently unmet clinical needs***
- ***potential impact on improving patients' quality of life***

- ***concerns about the widespread use of gene therapy making society less accepting of people who have mild disorders / genetic diseases or less able (e.g. intellect / height / strength etc.)***
- ***potential for non-therapeutic enhancement possibilities / eugenic social policies***

***Ethical Considerations***

- ***concerns over whether patients and their family members fully understand the risks associated with gene therapy trials***
- ***and whether they make informed decisions before participating in a trial***
- ***ref to Jesse Gelsinger / 1st death from gene therapy trial – he had a relatively mild form of a disease that was being successfully treated with medication and perhaps should not have been a gene therapy trial participant***

- ***while new gene therapy may have potential for tremendous patient benefit, yet many have unknown risks***
- ***raises questions over who should participate in a human trial – the terminally ill with no treatment options or anyone who understands the potential risk but is a willing patient***
- ***sponsors of human clinical trials must exercise responsibility in balancing experimental aims and ensuring that participants are not exposed to known risks / experimental products are as safe as possible***

- ***concerns about the protection of privacy and confidentiality of medical information of patients involved in clinical trials***
- ***which may have implications on (medical) insurance coverage / employability etc.***

- ***critics of gene therapy are concerned that gene therapy may be abused to alter human traits not associated with disease (such as using gene therapy to enhance a person's athletic ability, height or intellect)***
- ***many oppose to gene therapy on religious grounds, believing that altering genetic material (is against God's will / raises the specter of "playing God")***

[Total: 20]

**------ The End ------**